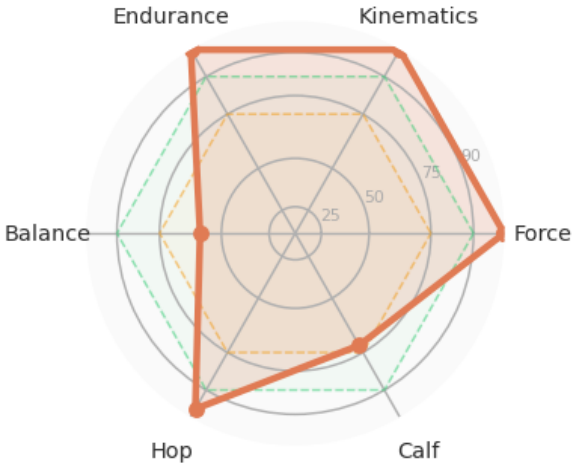
BYoung Physical Therapy

Patient: Jonathan Demay • **Date:** July 20, 2026
Surgical Side: Left • **Triple Hop:** Surg 22'0" / NS 22'8" / LSI 97.1% • **Heel Raise (reps):** Surg 18 / NS 25 / LSI 72.0%

Return to Sport Assessment
Achilles Repair

70 **RTS Score Progressing**
2 scored red flags

Return to Sport Profile



▲ STRENGTHS

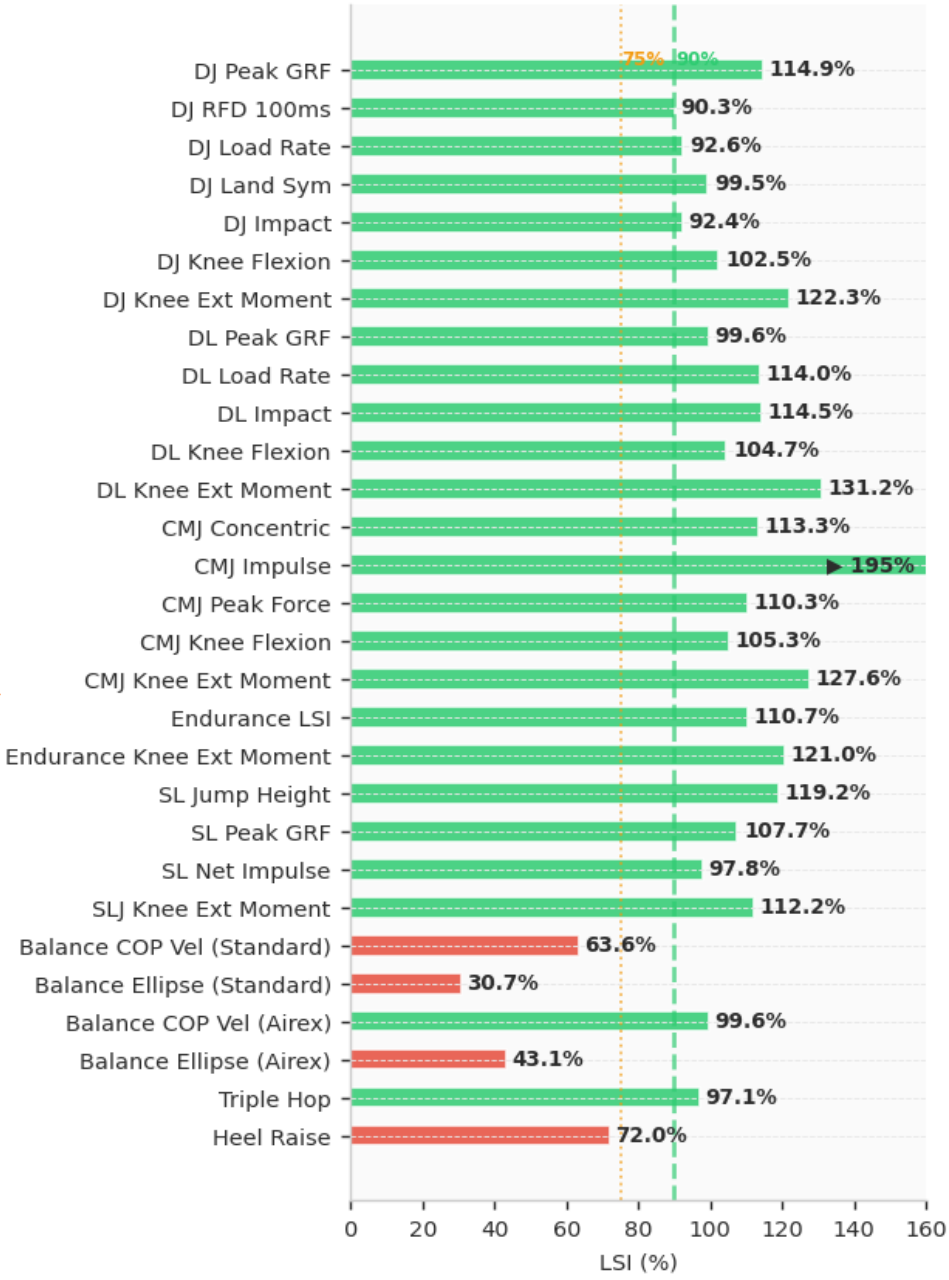
- Single-leg jump (100/100)
- DL Load rate sym. (100/100)
- DJ Peak GRF sym. (100/100)
- Endurance symmetry (100/100)

▼ DETRACTORS

- RSI (reactive strength) (31/100)
- Heel raise (calf capacity) (48/100)
- Fatigue drift (62/100)

Performance 78/100 → ×0.89 red-flag penalty (2 scored) → = 70/100

Limb Symmetry Index — All Tests



CLINICAL FLAGS

- Balance Ellipse (Standard): LSI 30.7% — significant asymmetry
- Balance Ellipse (Airex): LSI 43.1% — significant asymmetry
- Balance COP Vel (Standard): LSI 63.6% — significant asymmetry
- Heel Raise: LSI 72.0% — significant asymmetry

Achilles profile — rate-of-loading, reactive and calf-capacity metrics are weighted above peak force. Knee valgus / tibial rotation are shown for context, not scored. Ankle angle / moment signals were not found in the V3D export. Add L/R_Ankle_Angle and L/R_Ankle_Moment to the export pipeline for plantarflexor-specific metrics.

● ≥90% Symmetric ● ≥75% Progressing ● <75% Deficit LSI = Surgical ÷ Non-Surgical × 100 | Δ = Surgical – Non-Surgical (absolute difference; used for small-angle kinematics)

DROP JUMP

Metric	Surgical	Non-Surg	LSI / Δ
Peak GRF (N)	1359.9	1183.6	114.9%
Impulse (N·s)	733.7	683.0	107.4%
RFD 100ms (N/s)	11194.8	12404.2	90.3%
Loading Rate 50ms (N/s)	13434.0	14506.7	92.6%
Landing Sym 200ms (%) *(bilateral)*	—	—	99.5%
Impact Transient (N)	683.2	739.4	92.4%
Time-to-Peak (ms)	295.0	1249.0	—
Jump Height (in)	14.8	—	—
Contact Time (s)	0.710	—	—
RSI (m/s)	0.53	—	—
Peak Knee Flexion (°)	109.2	106.6	102.5%
Peak Knee Valgus (°)	9.2	2.0	Δ +7.2°
Peak Tibial IR (°)	-7.4	-1.7	Δ -5.7°
Tibial Rot Excursion (°)	6.1	7.8	Δ -1.6°
Peak Hip Adduction (°)	0.4	-1.6	Δ +2.0°
Peak Knee Ext Moment (Nm)	245.4	200.6	122.3%
AP COP Excursion (mm)	145.0	141.1	—

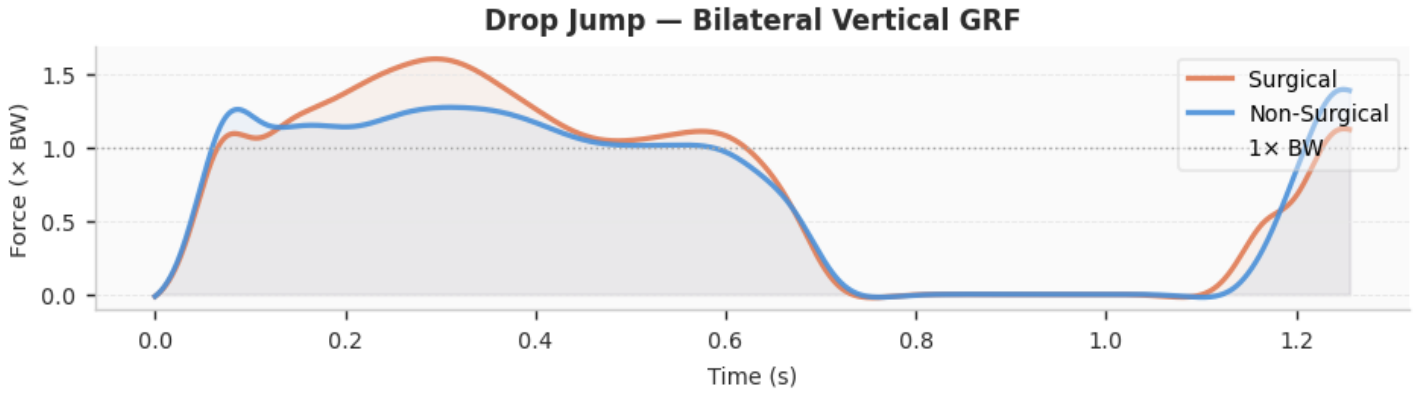
CLINICAL INTERPRETATION

RSI 0.53 m/s with 0.71 s contact (⚠ below RTS target ≥ 1.30). Long contact / low reactive strength — the surgical calf–Achilles unit is not yet tolerating fast stretch-shortening loads. Progress pogo → short-contact drop jumps.

Contact time 0.71 s — not a reactive (fast stretch-shortening) contact; target < 0.25 s.

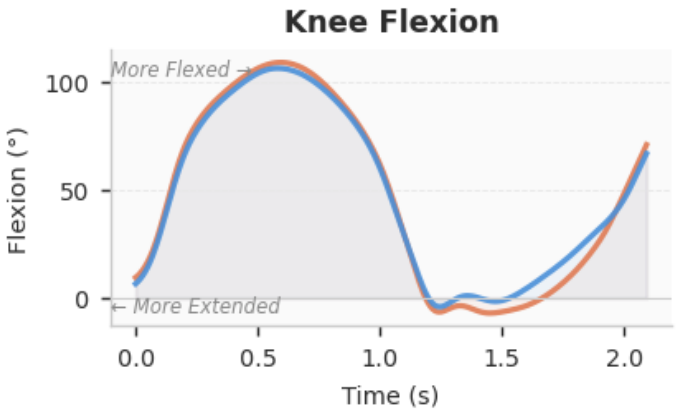
Landing symmetry 99% LSI: Good bilateral loading at contact.

Targets: RSI ≥ 1.30 m/s · Contact < 0.25 s · RFD & load rate LSI ≥ 85–90% · Landing LSI 90–110%.



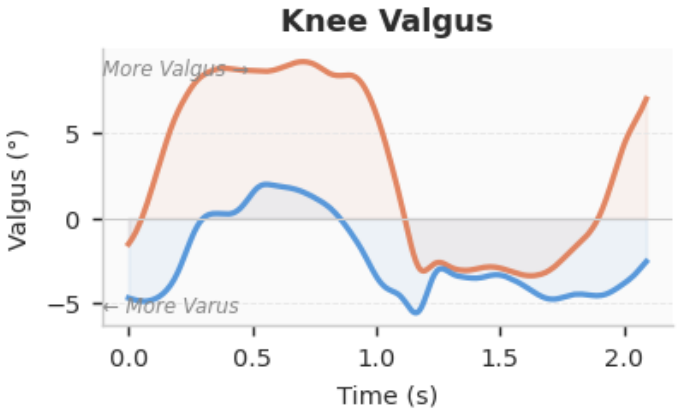
Bilateral GRF (normalised to BW). A slower, lower initial rise on the surgical side = reduced fast loading through the calf–Achilles unit.

Landing symmetry 99% at 200ms — symmetric bilateral loading.

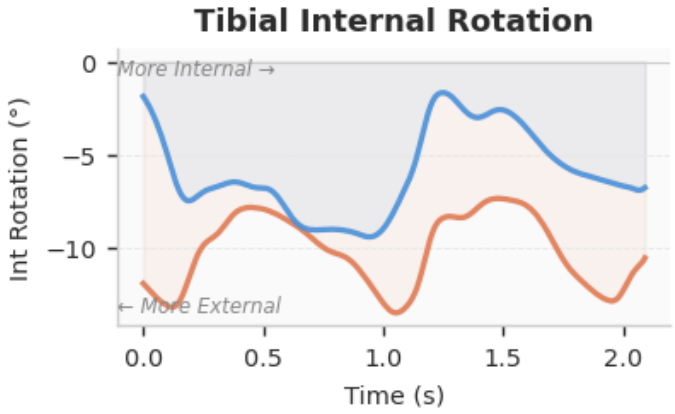


Knee flexion. Deeper knee bend on landing can be used to offload the ankle.

Surgical limb knee flexion depth is symmetric with the non-surgical side (LSI 103%) — meets return-to-sport threshold.



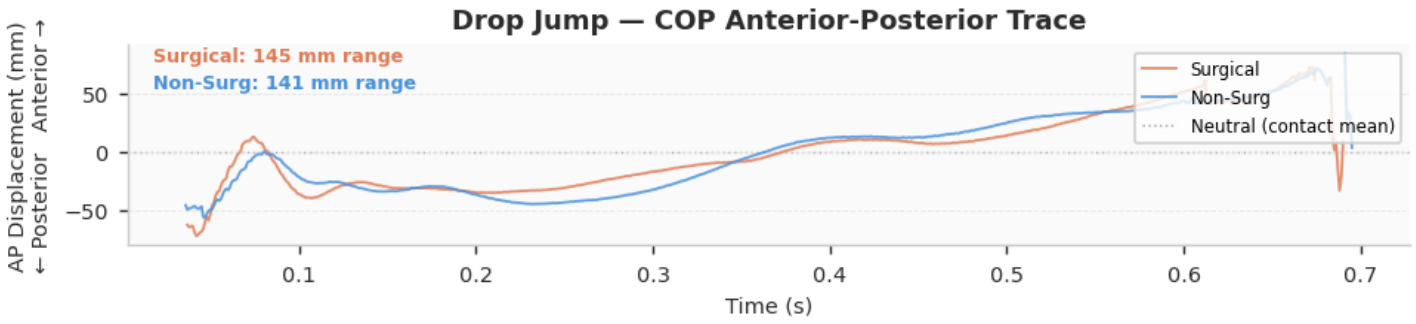
Knee valgus (context only for Achilles — not scored).



Tibial / foot rotation. A consistent surgical-side external-rotation offset can reflect toe-out (avoiding push-off through the forefoot) or a static calibration offset — confirm with video.

***Left panel:** Surgical knee valgus: 9.2° vs non-surgical 2.0° (Δ +7.2°) — side-to-side difference exceeds 6° (context only — not scored for this report type).*

***Right panel:** Surgical tibial IR: -7.4° vs non-surgical -1.7° (Δ -5.7°) — mild side-to-side asymmetry (context only — not scored for this report type).*



AP COP during the first contact (zero-centred). Posterior / heel bias on the surgical side = reduced forefoot loading through the Achilles.

AP COP excursion: surgical 145 mm vs non-surgical 141 mm — symmetric fore-aft weight shift.

Surgical limb bilateral landing symmetry is symmetric with the non-surgical side (LSI 99%) — meets return-to-sport threshold.

ABOUT THIS TEST

The Drop Jump is a reactive bilateral task: the athlete drops from a box and immediately jumps as high and as quickly as possible. It loads the calf–Achilles unit through a fast stretch-shortening cycle — the same demand as sprinting and cutting. After Achilles repair the common compensations are a longer ground contact, a slower and lower initial force rise on the surgical limb (reduced RFD / loading rate), a heel-biased landing, and shifting work to the knee and hip. The Reactive Strength Index (jump height ÷ contact time) summarises how well the athlete uses elastic energy; contact time itself shows whether the jump was truly reactive (< 0.25 s) or a slow, guarded rebound.

DROP LANDING

Metric	Surgical	Non-Surg	LSI / Δ
Peak GRF (N)	1093.6	1098.2	99.6%
Impulse (N·s)	782.1	656.3	119.2%
Loading Rate 50ms (N/s)	14122.0	12387.3	114.0%
Impact Transient (N)	757.1	661.0	114.5%
Peak Knee Flexion (°)	112.7	107.7	104.7%
Peak Knee Valgus (°)	9.8	1.4	Δ +8.4°
Peak Tibial IR (°)	-7.4	-1.3	Δ -6.1°
Tibial Rot Excursion (°)	5.7	8.2	Δ -2.5°
Peak Hip Adduction (°)	-5.5	-4.7	Δ -0.8°
Peak Knee Ext Moment (Nm)	198.7	151.4	131.2%
AP COP Excursion (mm)	104.9	88.2	—

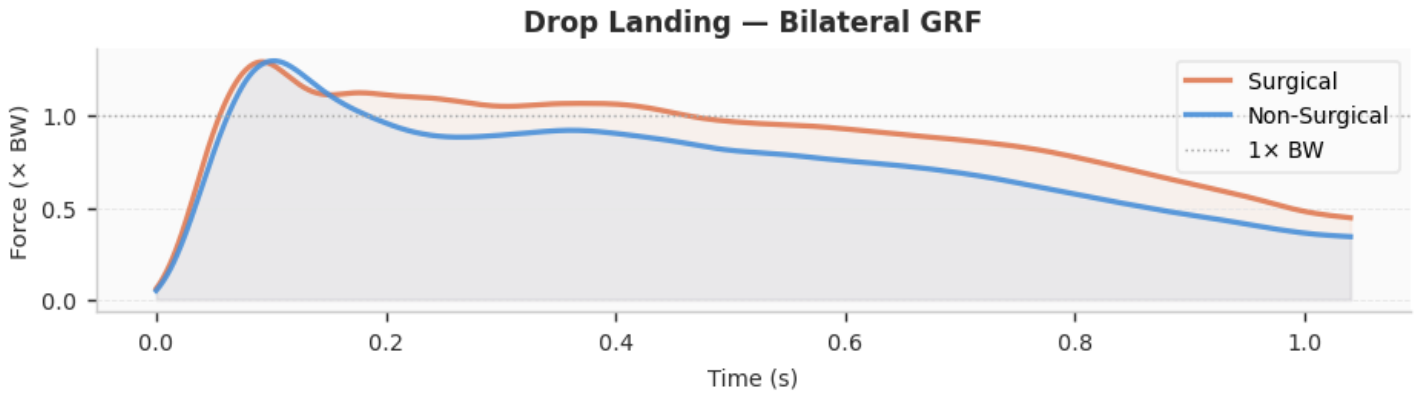
CLINICAL INTERPRETATION

Absorptive landing — assesses how quickly and symmetrically the surgical limb accepts load.

Load rate LSI 114%: Surgical limb loading rate 14% higher — NS side protective offloading.

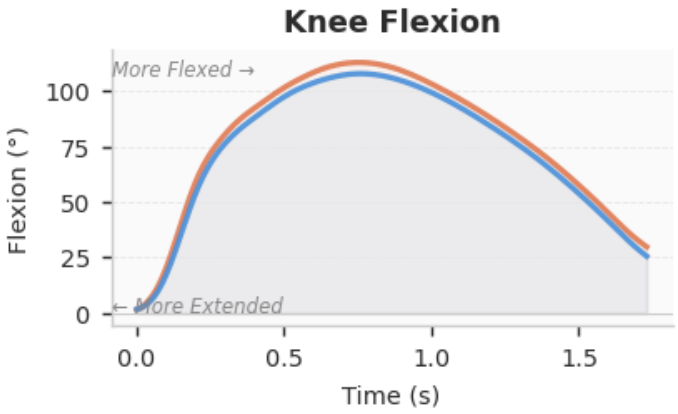
Impact transient LSI 115% — surgical limb initial impact 15% higher.

Targets: Load rate LSI ≥ 85–90% · Impact transient LSI 90–110% · Peak GRF LSI ≥ 90%.



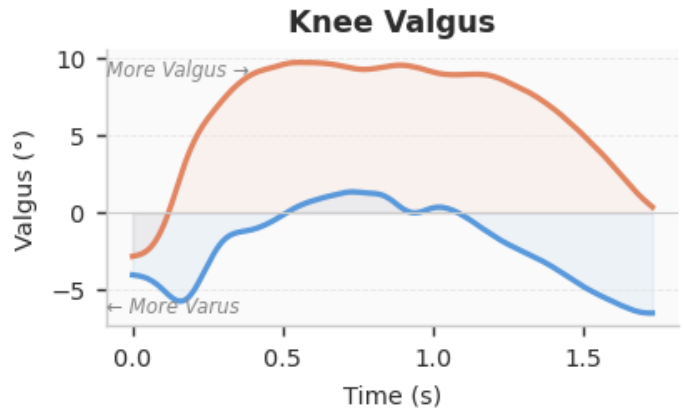
Bilateral GRF. Lower, later initial peak on the surgical side = slower load acceptance.

Peak GRF symmetry 100% — meets return-to-sport threshold.

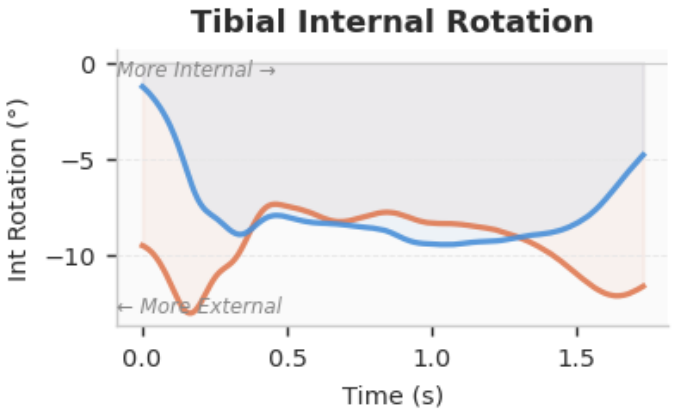


Knee flexion at landing. Extra knee bend can substitute for ankle absorption.

Surgical limb landing knee flexion depth is symmetric with the non-surgical side (LSI 105%) — meets return-to-sport threshold.



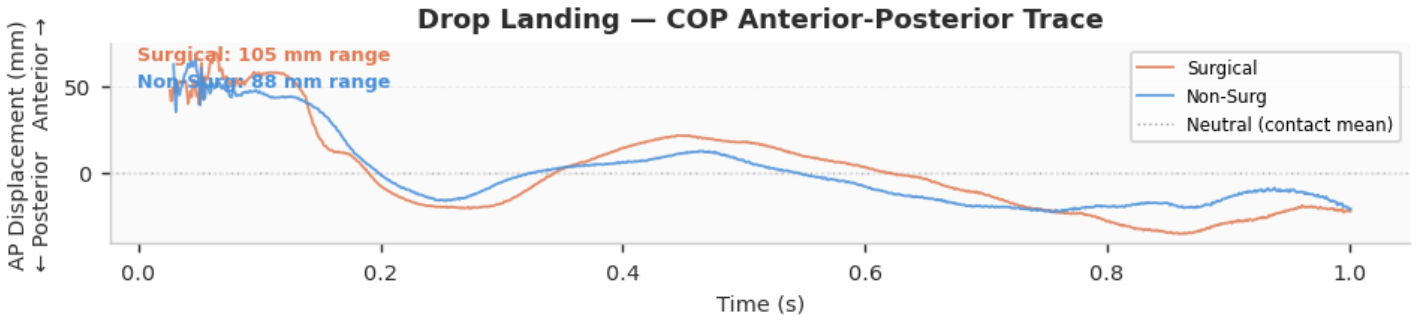
Knee valgus at landing (context only for Achilles — not scored).



Tibial / foot rotation. Surgical-side external rotation offset = possible toe-out strategy.

Left panel: *Surgical knee valgus: 9.8° vs non-surgical 1.4° (Δ +8.4°) — side-to-side difference exceeds 6° (context only — not scored for this report type).*

Right panel: *Surgical tibial IR: -7.4° vs non-surgical -1.3° (Δ -6.1°) — mild side-to-side asymmetry (context only — not scored for this report type).*



AP COP during landing contact (zero-centred). Posterior bias on surgical side = heel-dominant / reduced forefoot loading.

AP COP excursion: surgical 105 mm vs non-surgical 88 mm — more fore-aft travel on the surgical side — less controlled foot loading.

Surgical limb peak landing force absorption is symmetric with the non-surgical side (LSI 100%) — meets return-to-sport threshold.

ABOUT THIS TEST

The Drop Landing isolates load acceptance: the athlete drops from a box and sticks the landing without a re-jump. After Achilles repair the surgical limb often lands heel-first and accepts load more slowly, which shows up as a lower loading rate and impact transient rather than a lower peak force. Symmetric loading rate means the calf is eccentrically controlling the forefoot-to-heel transition on both sides.

MAX VERTICAL JUMP (CMJ)

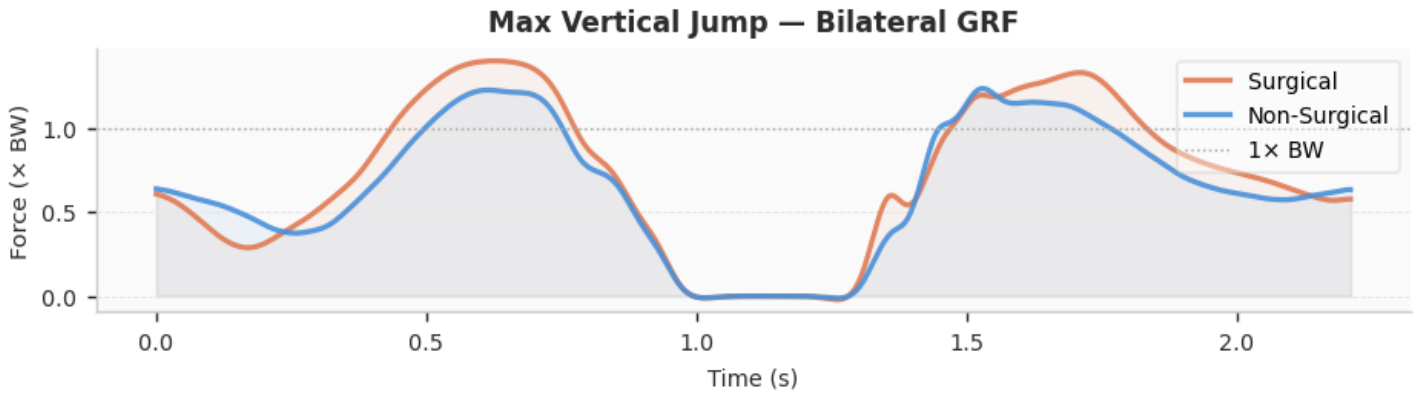
Metric	Surgical	Non-Surg	LSI / Δ
Peak Concentric GRF (N)	1186.4	1047.3	113.3%
Propulsion Impulse (N·s)	949.1	485.8	195.4%
Peak Force Surg / NS (N)	748.8	679.1	110.3%
Propulsion LSI (%) *(bilateral)*	—	—	195.4%
Jump Height (in)	16.0	—	—
Flight Time (s)	0.300	—	—
Unweighting Impulse Surg/NS	185.6	185.2	—
Peak Knee Flexion (°)	123.3	117.0	105.3%
Peak Knee Valgus (°)	10.3	2.0	Δ +8.3°
Peak Tibial IR (°)	-4.0	-1.0	Δ -3.0°
Tibial Rot Excursion (°)	7.8	6.8	Δ +1.0°
Peak Hip Adduction (°)	-1.1	-2.6	Δ +1.5°
Peak Knee Ext Moment (Nm)	236.1	185.1	127.6%
AP COP Excursion (mm)	166.0	155.3	—

CLINICAL INTERPRETATION

Bilateral CMJ — slow stretch-shortening; usually recovers before fast-SSC tasks after Achilles repair.

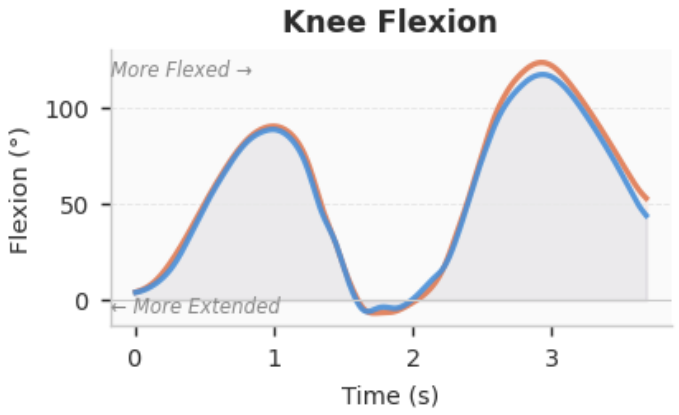
Peak force LSI 110%: Surgical limb generating 10% more force — possible NS compensation.

Propulsion LSI 195%. Targets: ≥ 90% propulsion, ≥ 90% peak force.



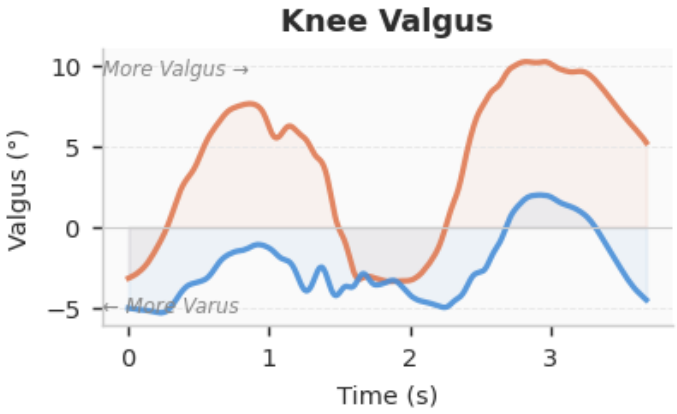
Full CMJ: unweighting → eccentric → propulsion → flight. Propulsive peaks should be symmetric.

Surgical limb propulsive impulse is symmetric with the non-surgical side (LSI 195%) — meets return-to-sport threshold.

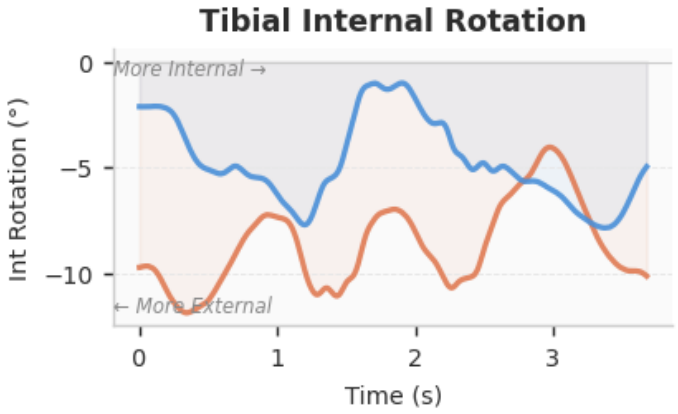


Knee flexion depth. Asymmetric countermovement depth = compensatory loading strategy.

Surgical limb countermovement knee flexion depth is symmetric with the non-surgical side (LSI 105%) — meets return-to-sport threshold.



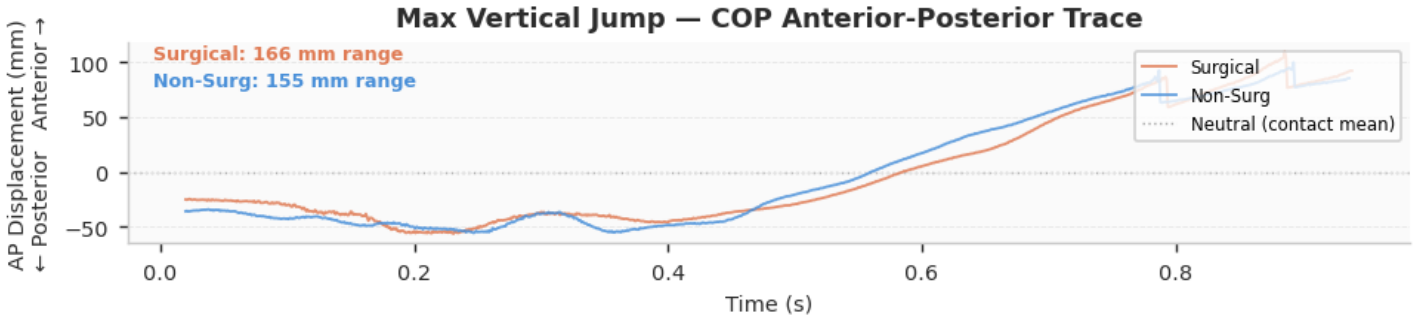
Knee valgus (context only for Achilles — not scored).



Tibial / foot rotation. Surgical-side external rotation offset = possible toe-out strategy.

***Left panel:** Surgical knee valgus: 10.3° vs non-surgical 2.0° (Δ +8.3°) — side-to-side difference exceeds 6° (context only — not scored for this report type).*

***Right panel:** Surgical tibial IR: -4.0° vs non-surgical -1.0° (Δ -3.0°) — side-to-side difference within acceptable range.*



AP COP during the propulsive contact (zero-centred). Less anterior travel on the surgical side = reduced push-off through the forefoot.

AP COP excursion: surgical 166 mm vs non-surgical 155 mm — symmetric fore-aft weight shift.

Surgical limb peak propulsive force is symmetric with the non-surgical side (LSI 110%) — meets return-to-sport threshold.

ABOUT THIS TEST

The Countermovement Jump is a bilateral power test with a relatively long, slow stretch-shortening cycle. Because the hip and knee can contribute most of the work, CMJ symmetry often returns before the calf has recovered — a symmetric CMJ does not clear an Achilles. Propulsive impulse and peak force LSI show whether the surgical limb is contributing to push-off.

ENDURANCE SQUAT (30s)

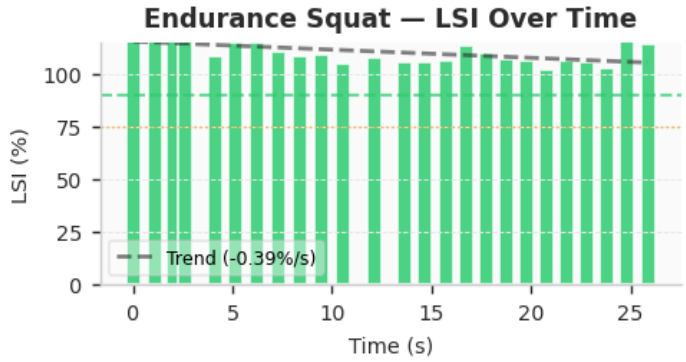
Metric	Surgical	Non-Surg	LSI / Δ
Mean Peak Force LSI (%) *(bilateral)*	—	—	110.7%
LSI First 10s (%) *(bilateral)*	—	—	116.2%
LSI Last 10s (%) *(bilateral)*	—	—	107.9%
Fatigue Drift (Δ%)	-8.3	—	—
Total Reps Detected	25.0	—	—
Mean Peak Force Surg (N)	1011.4	—	—
Mean Peak Force NS (N)	915.1	—	—
AP COP Excursion — Surg / NS (mm)	179.8	165.3	—
Peak Knee Ext Moment (Nm)	222.9	184.2	121.0%

CLINICAL INTERPRETATION

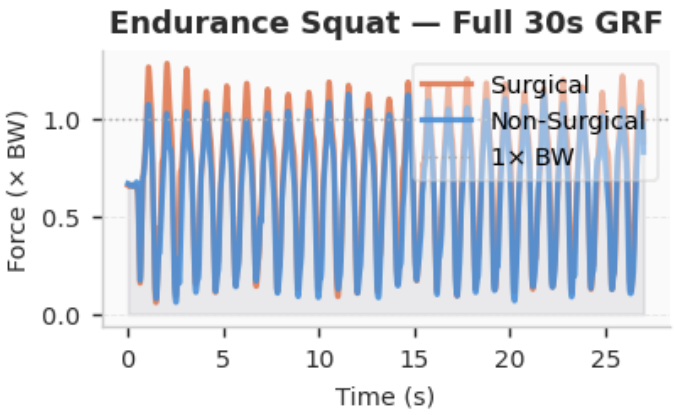
30s bilateral squat — knee-dominant endurance; does not directly load the calf.

Fatigue drift -8.3%: ⚠ LSI declined 8% — surgical limb offloads under prolonged effort.

Targets: Mean LSI ≥ 90% · Fatigue drift < 5%. Pair with the heel-raise test for calf endurance.

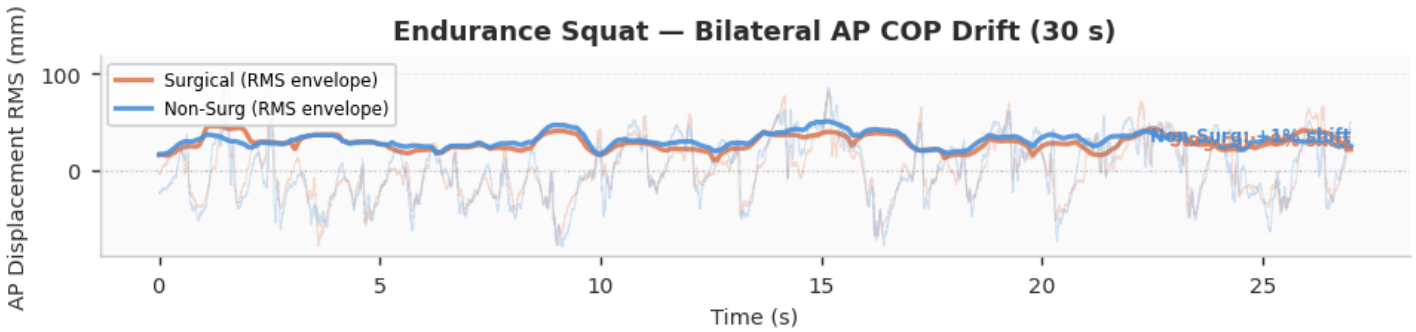


Per-rep LSI across 30s. Declining trend = neuromuscular fatigue breakdown.



Full 30s GRF trace. Force magnitude and rhythm across the set.

Left panel: *Surgical limb mean 30s force symmetry is symmetric with the non-surgical side (LSI 111%) — meets return-to-sport threshold.*



Bilateral AP COP drift across 30s (zero-centred, 1s RMS envelope). Surgical-side amplitude shrinking or trending posterior = offloading under fatigue. Diverging traces = progressive asymmetry with sustained effort.

Fatigue drift: -8.3% LSI change across 30s — surgical limb offloads progressively; 8% decline under fatigue.

Surgical limb mean force symmetry over 30 s is symmetric with the non-surgical side (LSI 111%) — meets return-to-sport threshold.

ABOUT THIS TEST

The 30-Second Endurance Squat assesses limb symmetry under sustained bilateral loading. It is knee- and hip-dominant, so for an Achilles repair it mainly checks for whole-limb offloading under fatigue; calf endurance is assessed with the single-leg heel raise to failure.

SINGLE-LEG VERTICAL JUMP

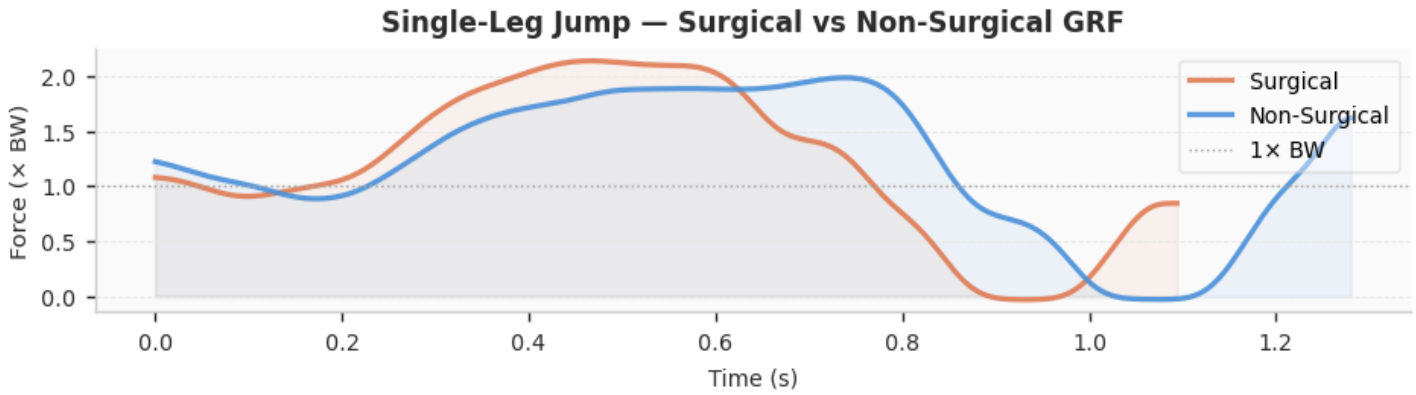
Metric	Surgical	Non-Surg	LSI / Δ
Jump Height (in)	7.1	5.9	119.2%
Peak GRF (N)	1805.1	1676.4	107.7%
Net Impulse (N·s)	309.2	316.1	97.8%
Contact Time (s)	0.89	1.02	—
Peak Knee Flexion (°)	62.5	60.6	—
Peak Knee Valgus (°)	8.4	-1.3	Δ +9.7°
Peak Tibial IR (°)	-10.3	-2.6	Δ -7.8°
Peak Hip Adduction (°)	12.3	4.2	Δ +8.0°
Peak Knee Ext Moment (Nm)	160.5	143.1	112.2%
AP COP Excursion (mm)	139.9	175.8	—

CLINICAL INTERPRETATION

Isolates each limb — the surgical calf must produce the final push-off without help.

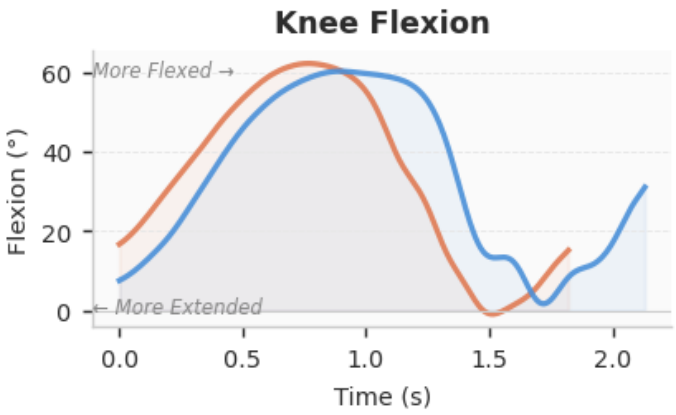
Jump height LSI 119%: ✓ Meets RTS threshold (≥ 90%).

Targets: Jump height LSI ≥ 90% · Net impulse LSI ≥ 90% · Heel-raise LSI ≥ 90%.

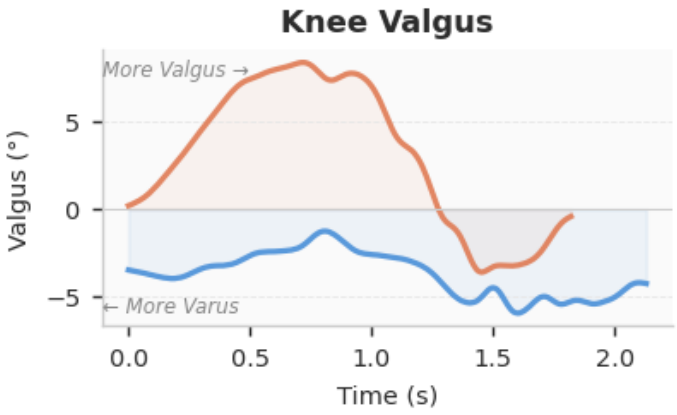


Per-limb GRF (separate trials). A later, drawn-out surgical peak = slow push-off.

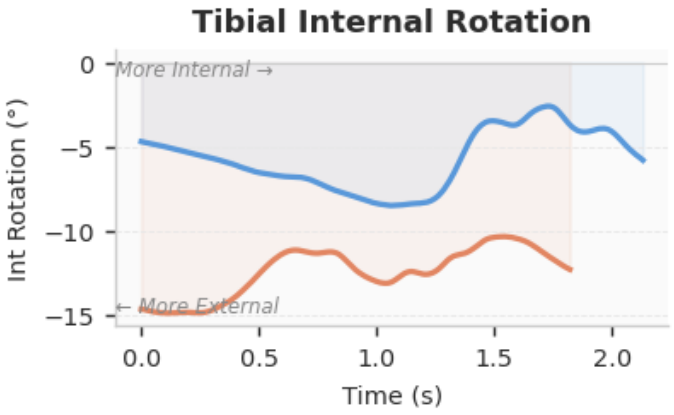
Surgical limb single-leg jump height is symmetric with the non-surgical side (LSI 119%) — meets return-to-sport threshold.



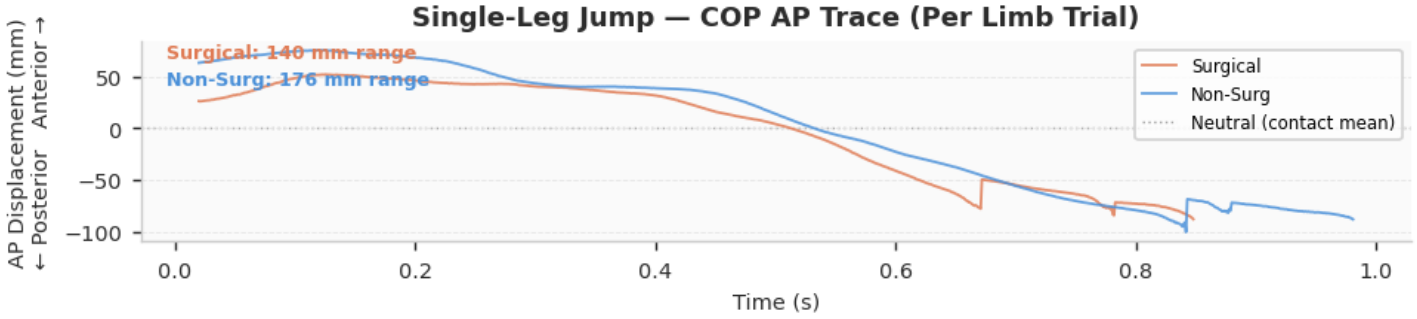
Knee flexion per limb. Reduced surgical depth = shortened, guarded countermovement.



Knee valgus per limb (context only for Achilles — not scored).



Tibial / foot rotation per limb. Surgical-side external rotation offset = possible toe-out.



AP COP during the take-off contact (zero-centred). Less anterior travel = limited push-off through the forefoot.

AP COP excursion: surgical 140 mm vs non-surgical 176 mm — less fore-aft travel on the surgical side — guarded or restricted loading.

Surgical limb single-leg jump height is symmetric with the non-surgical side (LSI 119%) — meets return-to-sport threshold.

ABOUT THIS TEST

The Single-Leg Vertical Jump isolates each limb so the uninvolved side cannot compensate. Final push-off depends on the plantarflexors, so after Achilles repair single-leg height is one of the more sensitive power measures. Compare jump height with net impulse: if impulse is near-symmetric but height is not, the athlete is producing force slowly over a longer contact rather than quickly through the calf.

BALANCE — STANDARD (EYES CLOSED, FIRM SURFACE)

Metric	Surgical	Non-Surg	LSI / Δ
COP Mean Velocity (mm/s)	157.7	100.2	63.6%
COP Ellipse Area (mm²)	14078.4	4319.7	30.7%
Total Excursion (mm)	2207.3	1403.2	—
AP Range (mm)	106.0	89.7	—
ML Range (mm)	278.9	53.1	—
RMS Displacement (mm)	45.7	22.2	—

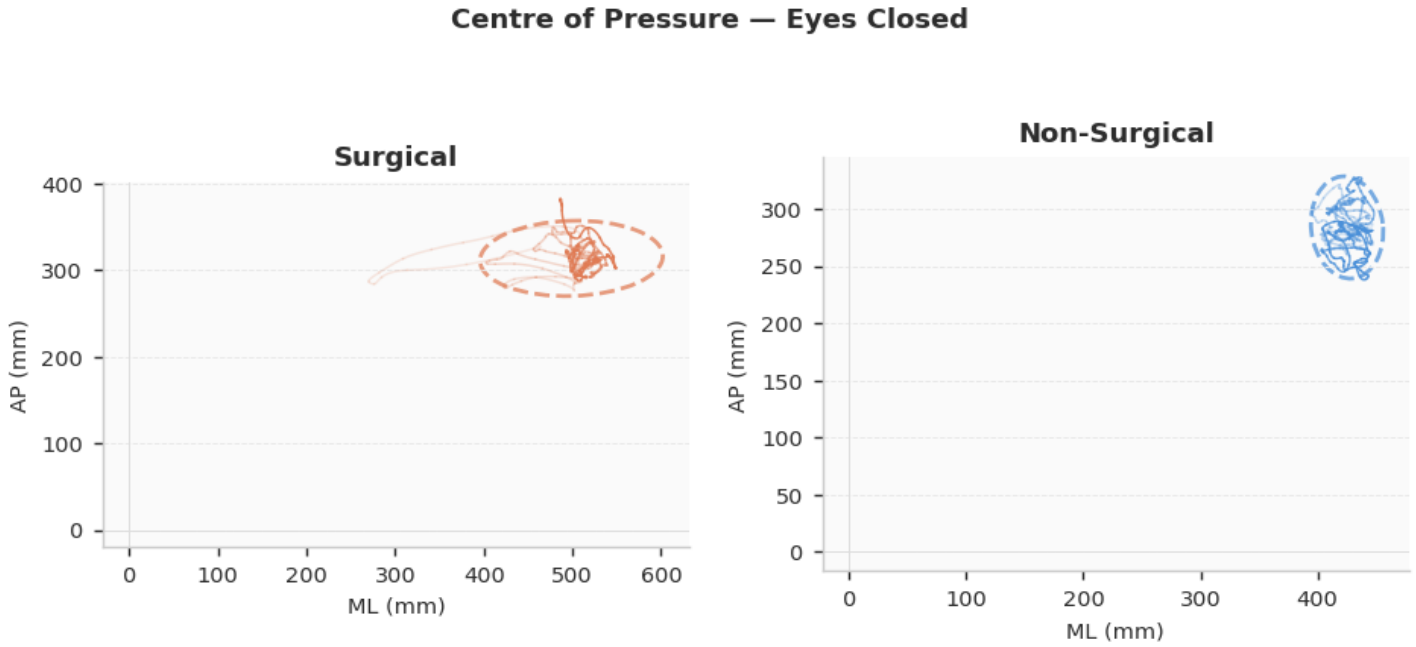
CLINICAL INTERPRETATION

Eyes-closed single-leg stance on firm surface. COP velocity is the most sensitive metric — higher speed = more corrective activity = reduced proprioception.

Surgical 158 mm/s vs Non-Surgical 100 mm/s — Surgical limb less stable. Proprioceptive retraining indicated.

COP Velocity LSI 64% (NS÷Surg×100; below 100% = surgical side sways more). LSI target ≥ 90%. COP low-pass filtered at 10 Hz.

⚠ **Data check:** COP range on the surgical trial exceeds 150 mm — likely a foot shift or touchdown rather than sway. Review / re-collect before relying on this page.



COP trajectory (10 Hz low-pass) — eyes closed. Tighter ellipse = steadier stance. 95% confidence ellipse shown.

ABOUT THIS TEST

Eyes-closed single-leg stance removes visual feedback, so balance depends on ankle and foot proprioception and on the calf's ability to make fast small corrections. Higher COP velocity means more corrective activity. After Achilles repair, reduced calf strength and altered tendon stiffness commonly increase sway on the surgical limb. Limb symmetry (LSI ≥ 85–90%) is the primary criterion.

BALANCE — AIREX PAD (EYES CLOSED, UNSTABLE SURFACE)

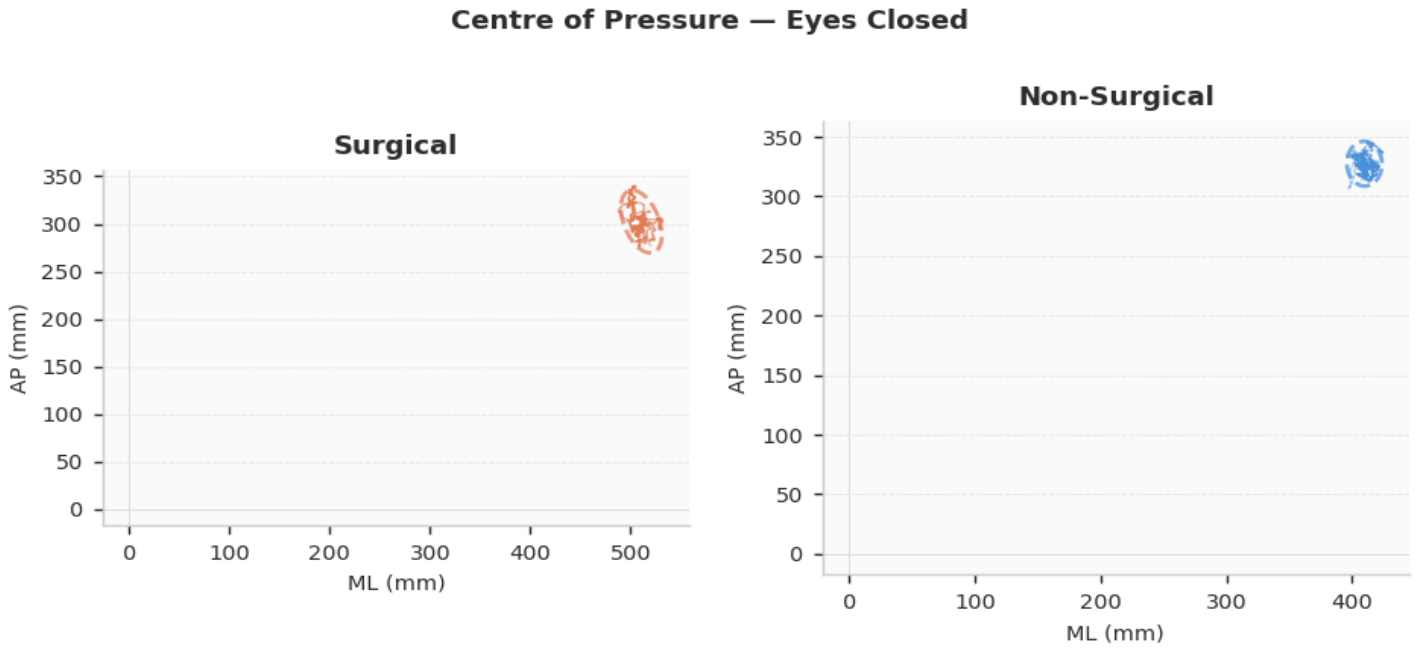
Metric	Surgical	Non-Surg	LSI / Δ
COP Mean Velocity (mm/s)	55.5	55.2	99.6%
COP Ellipse Area (mm²)	1904.4	821.5	43.1%
Total Excursion (mm)	776.6	773.1	—
AP Range (mm)	63.3	39.0	—
ML Range (mm)	43.9	27.9	—
RMS Displacement (mm)	15.9	9.5	—

CLINICAL INTERPRETATION

Eyes-closed single-leg stance on Airex foam pad. Higher COP values are expected — the unstable surface challenges proprioceptive and neuromuscular systems. Symmetry between limbs (LSI ≥ 85%) is the key target.

Surgical 55 mm/s vs Non-Surgical 55 mm/s — COP velocity comparable between limbs.

COP Velocity LSI 100% (NS÷Surg×100; below 100% = surgical side sways more). LSI target ≥ 90%. COP low-pass filtered at 10 Hz.



COP trajectory (10 Hz low-pass) — eyes closed. Tighter ellipse = steadier stance. 95% confidence ellipse shown.

ABOUT THIS TEST

Eyes-closed single-leg stance removes visual feedback, so balance depends on ankle and foot proprioception and on the calf's ability to make fast small corrections. Higher COP velocity means more corrective activity. After Achilles repair, reduced calf strength and altered tendon stiffness commonly increase sway on the surgical limb. Limb symmetry (LSI ≥ 85–90%) is the primary criterion.